

Rational Functions and Asymptotes (2.7)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $(1, -1)$ # _____ State the vertical asymptote of $f(x) = \frac{1}{x-2}$.ANSWER: $x = 1$ # _____ State the vertical asymptote of $f(x) = \frac{(x-1)(x+2)}{(x-1)(x-4)}$.ANSWER: $y = 0$ # _____ State the horizontal asymptote of $f(x) = \frac{2x^2}{x^2-1}$.ANSWER: $x = -3, x = 3$ # _____ State the horizontal asymptote of $f(x) = \frac{3x^2+1}{x^2}$.

ANSWER: $x = 2$ # _____ State the horizontal asymptote of $f(x) = \frac{1}{x}$.	ANSWER: none # _____ State the vertical asymptotes of $f(x) = \frac{1}{x^2 - 9}$.
ANSWER: $x \neq -3$ # _____ Find the x-location of the hole in $f(x) = \frac{(x-1)(x+2)}{(x-1)(x-4)}$.	ANSWER: $y = 3$ # _____ Give the coordinates of the hole in $f(x) = \frac{(x-1)(x+2)}{(x-1)(x-4)}$.
ANSWER: $y = 2$ # _____ State the domain restriction of $f(x) = \frac{1}{x+3}$.	ANSWER: $x = 4$ # _____ State the horizontal asymptote of $f(x) = \frac{x^2 + 1}{x}$.